

Design and Implementation of a Database Management System for Hostel Accommodation and Student Services Management System

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Problem Identification

Existing Problems



Record Keeping

Manual hostel records are difficult to maintain and student transfer history may be lost

Bed Allocation

Duplicate bed allocation may occur and room capacity may be exceeded

Tracking & Monitoring

Difficult to track vacant and occupied beds; fee payments and dues are difficult to monitor

Services & Requests

Complaints and maintenance requests may be missed; visitor records require systematic management

Aim & Objectives

Aim

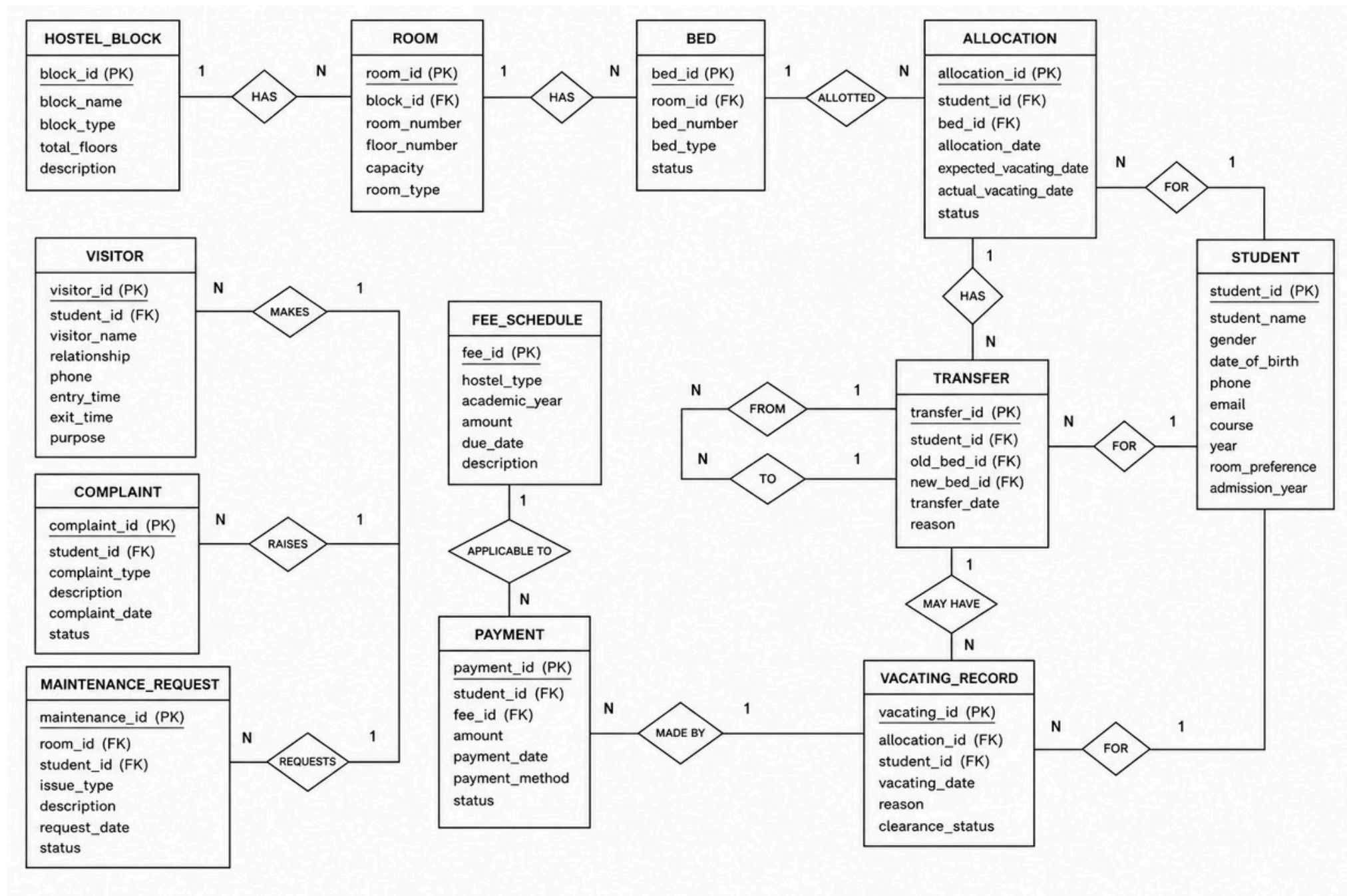
To design and implement a relational database system for efficient management of hostel accommodation and student services.



Objectives

- Manage student and hostel information
- Manage blocks, rooms and beds
- Prevent duplicate bed allocation
- Track student transfers and vacating
- Manage hostel fees and payments
- Maintain visitor records
- Track complaints and maintenance
- Generate occupancy and vacancy reports
- Maintain data integrity and consistency

ER DIAGRAM:



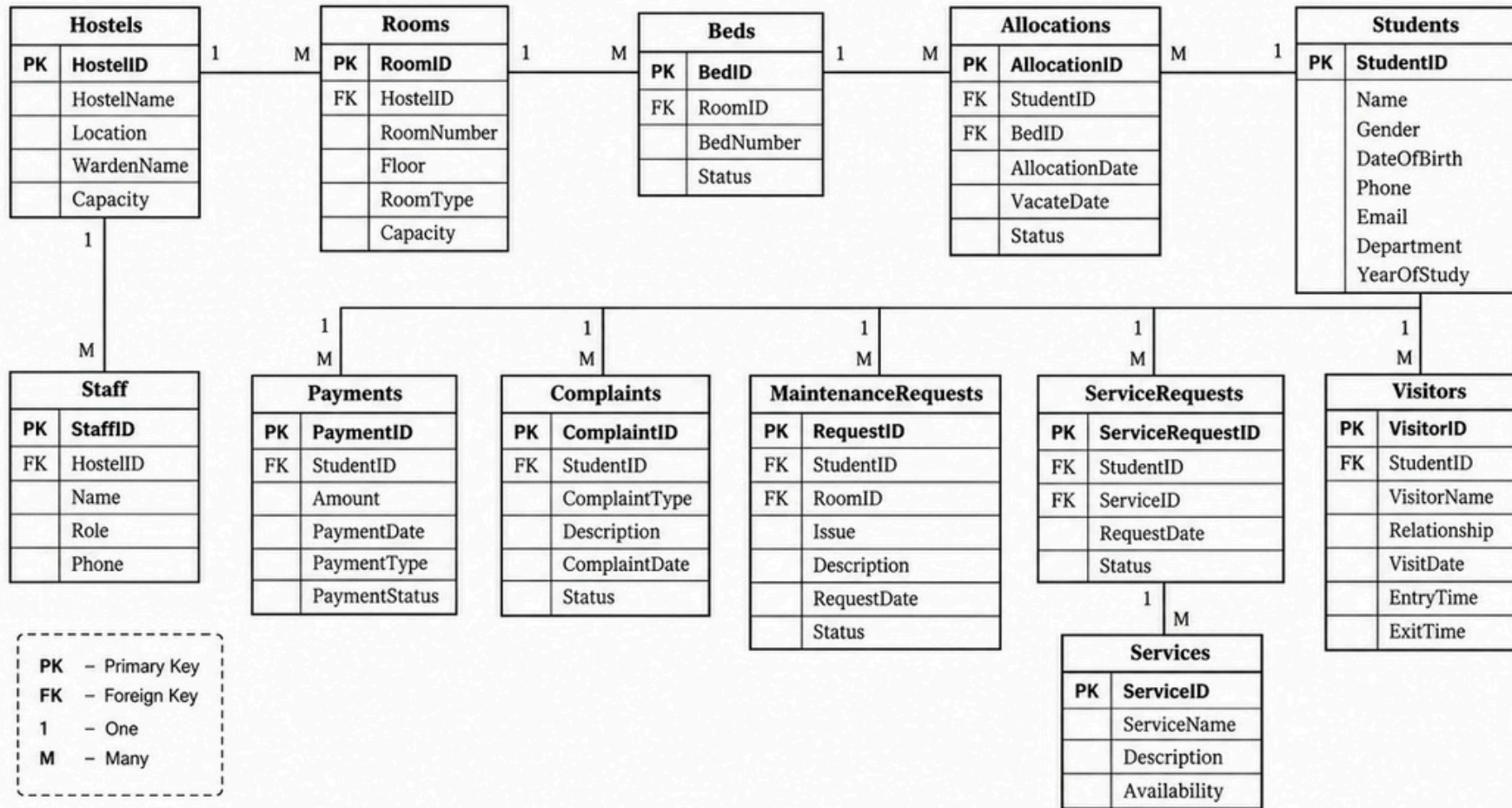
Initial Relational Schema

Proposed Tables

#	Table Name	Key Attributes
1	HOSTEL_BLOCK	block_id (PK)
2	ROOM	room_id (PK), block_id (FK)
3	BED	bed_id (PK), room_id (FK)
4	STUDENT	student_id (PK)
5	ALLOCATION	allocation_id (PK), student_id (FK), bed_id (FK)
6	TRANSFER	transfer_id (PK), student_id (FK)
7	FEE_SCHEDULE	fee_id (PK)
8	PAYMENT	payment_id (PK), student_id (FK), fee_id (FK)
9	VISITOR	visitor_id (PK), student_id (FK)
10	COMPLAINT	complaint_id (PK), student_id (FK)
11	MAINTENANCE_REQUEST	maintenance_id (PK), student_id (FK), room_id (FK)
12	VACATING_RECORD	vacating_id (PK), allocation_id (FK), student_id (FK)

Database Design

The database is designed using a relational model. It consists of 12 tables that store information about students, hostels, rooms, beds, accommodation, payments, complaints, maintenance requests, services, staff and visitors. The tables are related using primary keys (PK) and foreign keys (FK) to maintain data consistency and reduce redundancy.



Integrity Constraints

	PRIMARY KEY Unique identification of records
	FOREIGN KEY Maintains table relationships
	NOT NULL Prevents missing required values
	UNIQUE Prevents duplicate values
	CHECK Validates data values
	DEFAULT Provides default values

Rules

One active resident per bed	One active allocation per student
Room capacity cannot be exceeded	Allocation dates must be valid
Vacating date must be valid	Payment amount must be positive
Payment cannot exceed outstanding fees	

Expected Outcomes

Expected Outcomes

Centralized hostel management database

Accurate room and bed availability

Prevention of duplicate allocations

Efficient fee and payment tracking

Complete student allocation history

Organized visitor records

Faster complaint resolution

Efficient maintenance tracking

Accurate occupancy and vacancy reports

Improved data integrity and reliability

Future Implementation

MySQL Database+PythonApplication

MySQL
Relational Database
Engine

Python
Application Layer

Thank You